

MedTrack: Hospital Operations & Resource Utilization Dashboard

Module 8: Documentation and Project Delivery

Final Project Documentation | Version 1.0

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1. Project Overview

MedTrack: Hospital Operations & Resource Utilization Dashboard is a data visualization and analytics project developed to provide a consolidated view of hospital operations, patient activity, departmental performance, and resource utilization. The project transforms healthcare operational datasets into interactive Tableau dashboards that allow users to examine key performance indicators, identify operational patterns, compare departments, and understand areas that may require management attention.

Hospitals generate large volumes of operational data related to patient admissions, discharges, departments, services, bed capacity, occupancy, length of stay, readmissions, staffing, and resource usage. When these datasets are examined independently in spreadsheets or raw tables, it becomes difficult to identify relationships and trends quickly. MedTrack addresses this problem by organizing the available data into a structured analytical workflow and presenting the resulting information through interactive visualizations.

The project follows a complete data analytics workflow beginning with data collection and dataset integration, followed by data cleaning and transformation, healthcare KPI engineering, dashboard development, dashboard integration, and testing and validation. The processed data is converted into Tableau-ready datasets containing the measures required for hospital performance and operational analysis.

The solution is organized into four interconnected dashboards:

1. Hospital Overview

Provides an executive-level summary of overall hospital performance. It focuses on major KPIs such as total patients, occupancy rate, bed utilization rate, average length of stay, readmission rate, and department efficiency. Supporting visualizations allow users to examine admission distribution, departmental performance, occupancy, length of stay, and readmission patterns.

2. Patient Flow

Focuses on the movement of patients through hospital operations. It analyzes admission and discharge trends, patient volume across departments, average length of stay, and admitted versus refused patients. The dashboard helps identify changes in patient demand and variations in patient stay across departments and time periods.

3. Department Analytics

Provides a comparative view of individual hospital departments. It enables users to evaluate department efficiency, occupancy, bed utilization, patient/service volume, and readmission performance. This allows differences between departments to be identified rather than treating the hospital as a single operational unit.

4. Resource Analysis / Resource Utilization

Examines the utilization and availability of hospital resources, with particular emphasis on bed capacity and department-level resource performance. It combines occupancy, bed utilization, total bed capacity, and department efficiency to help identify differences in resource usage and potential areas for operational improvement.

The dashboards are designed to work together as a single analytical dashboard suite rather than as four independent reports. Navigation controls allow users to move between the Hospital Overview, Patient Flow, Department Analytics, and Resource Analysis dashboards. Consistent visual design, KPI definitions, filters, and dashboard interactions are used across the suite to provide a unified user experience.

The primary intended users are hospital administrators, healthcare managers, operational teams, and other decision-makers who require a summarized view of hospital activity and performance. The dashboards are designed to support operational monitoring and data-driven decision-making rather than clinical diagnosis or direct patient treatment decisions.

The final project deliverable is an integrated Tableau workbook containing the four dashboards, supported by cleaned and engineered datasets, KPI definitions, data-processing scripts, testing documentation, healthcare operations methodology, and project documentation. The project specification identifies these four dashboards and the complete workflow from healthcare data collection through documentation and delivery.

Project Workflow

Data Collection → Data Cleaning & Transformation → KPI Engineering → Dashboard Development → Dashboard Integration → Testing & Validation → Documentation & Delivery

This workflow ensures that MedTrack is not simply a collection of visualizations, but a complete healthcare operations analytics solution with documented data preparation, defined KPIs, interactive dashboards, validation procedures, and a structured final delivery.

2. Project Objectives

The primary objective of MedTrack is to transform hospital operational data into an interactive analytical solution that enables hospital stakeholders to understand performance, patient flow, departmental efficiency, and resource utilization through a unified Tableau dashboard suite.

The specific objectives are:

2.1 Monitor Hospital Performance

Track important hospital-level Key Performance Indicators (KPIs) such as:

- Total patient admissions
- Occupancy rate
- Bed utilization rate
- Average Length of Stay (LOS)
- Readmission rate
- Department efficiency score

These indicators provide a consolidated view of the hospital's operational performance.

2.2 Analyze Patient Flow

Analyze the movement of patients through different stages of hospital operations by examining:

- Patient admissions
- Patient discharges
- Monthly admission and discharge trends

- Patient volume across departments
- Average Length of Stay
- Patients admitted versus patients refused

This helps identify changes in patient demand, workload patterns, and variations in patient stay.

2.3 Evaluate Departmental Performance

Compare different hospital departments using operational and performance indicators such as:

- Patient/service volume
- Department efficiency
- Occupancy
- Bed utilization
- Readmission rate
- Average Length of Stay

The objective is to identify differences in departmental performance and highlight areas that may require further operational attention.

2.4 Analyze Hospital Resource Utilization

Evaluate how available hospital resources are being utilized, particularly:

- Total bed capacity
- Bed utilization

- Occupancy
- Department-level resource usage
- Relationship between occupancy and departmental efficiency

This provides a basis for understanding capacity usage and identifying potential opportunities for better resource allocation.

2.5 Provide Interactive Data Visualization

Develop interactive Tableau dashboards that allow users to:

- Filter and compare departments
- Explore operational trends
- View KPI details
- Interact with charts and tables
- Navigate between the four dashboards
- Examine hospital performance from different analytical perspectives

2.6 Support Data-Driven Operational Decisions

Convert raw healthcare operational data into understandable visual insights that can assist hospital administrators and operational stakeholders in identifying:

- High-demand periods
- Departments with high patient volumes
- Differences in departmental efficiency

- Length-of-stay patterns
- Resource utilization patterns
- Areas requiring further investigation

The objective is not to make clinical diagnoses or treatment decisions, but to provide an analytical foundation for monitoring hospital operations and supporting informed management decisions.

Overall Objective

To develop an integrated, interactive Tableau dashboard suite that transforms hospital operational data into meaningful KPIs and visual insights, enabling systematic analysis of hospital performance, patient flow, departmental efficiency, and resource utilization.

3.Dataset Sources

The MedTrack project uses multiple healthcare-related datasets to support patient-level analysis, department-level operational analysis, staffing analysis, and readmission analysis. The datasets were selected and prepared to provide the underlying information required for the project's KPI calculations and Tableau dashboards.

3.1 Final Source Datasets

The final documented datasets used in the project are:

Dataset	Primary Purpose	Key Analytical Use
patients_clean.csv	Patient-level information	Admissions, discharges, Length of Stay, patient-level analysis
services_weekly.csv	Weekly hospital service/department information	Department performance, admissions, occupancy, resource analysis
staff.csv	Staff information	Staff-related analysis and resource planning
staff_schedule.csv	Staff scheduling information	Staffing and allocation analysis
hospital_readmission_dataset.csv	Readmission information	Readmission analysis and outcome-related KPIs

3.2 Dataset Source Selection

During the dataset selection stage, several publicly available healthcare data repositories were explored, including:

- Kaggle
- UCI Machine Learning Repository

- HealthData.gov
- CMS Open Data

The project documentation identifies Kaggle as the selected source for the final operational datasets because the available datasets provided a sufficiently consistent representation of a single-hospital operational workflow. This consistency was important for combining patient, service, and operational information without introducing unnecessary differences in data structure or hospital context.

3.3 Role of Each Dataset

Patient Dataset

patients_clean.csv provides patient-level information required for analyzing patient admissions and discharges. It forms the basis for patient-flow analysis and supports the calculation of Length of Stay.

Weekly Services Dataset

services_weekly.csv contains weekly service or department-level operational information. It supports analysis of patient volume, occupancy, department performance, and resource-related indicators.

Staff Dataset

staff.csv provides information about hospital staff. It was included to support staffing-related analysis and the broader resource-utilization scope of the project.

Staff Schedule Dataset

staff_schedule.csv contains scheduling information that can be used to understand staff allocation and availability across operational periods.

Readmission Dataset

hospital_readmission_dataset.csv provides information required for readmission analysis. It supports the Readmission Rate KPI and the readmission-related visualizations included in the dashboard suite.

3.4 Dataset Preparation

The source datasets were not used directly in their raw form. They were subjected to data cleaning and transformation before being used for KPI engineering and Tableau visualization. The preparation process included:

- Reviewing dataset structure and available fields
- Identifying missing and duplicate records
- Standardizing column names
- Standardizing department/service names
- Converting fields to appropriate data types
- Deriving analytical fields such as Length of Stay
- Preparing datasets for KPI calculation
- Producing Tableau-ready datasets

The project specification identifies data collection, cleaning, transformation, and creation of Tableau-ready datasets as dedicated stages of the project workflow.

3.5 Data Flow

The datasets feed into the project through the following process:

Source Healthcare Datasets
↓
Data Cleaning & Transformation
↓
Dataset Integration
↓
KPI Engineering
↓
Tableau-Ready Data
↓
Four MedTrack Dashboards

This dataset preparation and integration stage provides the foundation for the KPI calculations and visual analytics used throughout the MedTrack dashboard suite.

4. Data Preparation and Quality

Data preparation was an essential stage of the MedTrack project because the quality and consistency of the underlying healthcare data directly affect the reliability of the KPIs and dashboard visualizations. The objective of this stage was to convert the collected datasets into clean, consistent, and analysis-ready datasets suitable for KPI engineering and Tableau visualization.

4.1 Data Preparation Workflow

The data preparation process followed a structured sequence:

Data Profiling → Quality Assessment → Data Cleaning → Standardization → Feature Engineering → Tableau-Ready Datasets

Each stage served a specific purpose in preparing the healthcare data for analysis.

4.2 Data Profiling

The datasets were initially examined to understand their structure and content. The profiling stage focused on:

- Identifying the available datasets and their respective purposes

- Reviewing the number and names of columns
- Understanding data types
- Identifying categorical and numerical variables
- Examining date fields required for patient-flow analysis
- Understanding the grain and structure of each dataset

This step established the foundation for subsequent cleaning and integration.

4.3 Data Quality Assessment

The datasets were assessed for common data-quality issues before being used for analysis. The checks included:

- Schema inspection: verifying the structure and expected fields of each dataset.
- Missing-value analysis: identifying fields containing null or incomplete values.
- Duplicate detection: identifying duplicate records that could distort patient counts or operational metrics.
- Data-type verification: ensuring that numerical, categorical, and date fields were stored in appropriate formats.
- Categorical consistency: checking whether department and service names were represented consistently.
- Numerical validation: checking numerical fields for invalid or unexpected values.
- Whitespace trimming: removing unnecessary leading or trailing spaces from text values.
- Column-name standardization: creating consistent and readable field names across datasets.

4.4 Data Cleaning

After identifying data-quality issues, the datasets were cleaned and transformed into a consistent analytical format.

The cleaning process included:

- Removing duplicate records where identified
- Handling missing or incomplete values
- Standardizing department/service names
- Correcting data types
- Normalizing healthcare-related indicators
- Removing unnecessary whitespace
- Standardizing column names
- Preparing the cleaned datasets for integration and KPI engineering

The project specification specifically identifies duplicate removal, missing-data handling, department-name standardization, healthcare-indicator normalization, and creation of Tableau-ready datasets as part of the cleaning stage.

4.5 Feature Engineering

Additional analytical fields were derived from the cleaned data where required.

A key derived metric was Length of Stay (LOS), calculated from the patient's arrival/admission date and departure/discharge date.

Conceptually:

Length of Stay = Departure/Discharge Date – Arrival/Admission Date

This derived field enables analysis of:

- Average patient stay
- Length of stay by department
- Monthly LOS patterns
- Department-level differences in patient stay

Other KPI-related fields were prepared during the KPI engineering stage for use in the final Tableau dashboards.

4.6 Standardization

Department and service names were standardized to ensure that the same department was not treated as multiple categories because of differences in spelling, capitalization, or formatting.

For example, inconsistent representations such as:

"General Medicine", "general medicine", and " General Medicine "

should be treated as the same department after standardization.

This is particularly important for Tableau because inconsistent category values can produce incorrect department counts, duplicated categories, and misleading comparisons.

4.7 Tableau-Ready Dataset

After cleaning, standardization, and feature engineering, the datasets were prepared for KPI calculation and Tableau visualization.

The resulting analytical data supports:

- Hospital-level KPI calculation
- Patient-flow analysis
- Department comparisons
- Occupancy analysis
- Bed-utilization analysis
- Readmission analysis
- Length-of-Stay analysis
- Resource-utilization analysis

The project specification identifies Tableau-ready datasets as the output of the data-cleaning stage and hospital_final_dataset.xlsx as the deliverable from the subsequent KPI-engineering stage.

4.8 Data Quality Targets

The project established quality targets for the preparation stage:

- Dataset completeness: greater than 95%
- Missing values after cleaning: less than 2%
- KPI accuracy during validation: greater than 95%

These targets provide measurable criteria for evaluating the quality of the data and the reliability of the resulting dashboard analytics.

4.9 Final Data Preparation Outcome

The outcome of this stage was a set of cleaned and standardized healthcare datasets that could be reliably used for KPI engineering and Tableau dashboard development. The preparation process established consistent data structures, reduced data-quality issues, and created the analytical fields required for the four-dashboard MedTrack solution.

5. Data Integration and Architecture

Data integration is a critical component of the MedTrack project because the analysis combines information from datasets that represent different aspects of hospital operations. The integration approach was designed to preserve the original structure and meaning of each dataset while making the required information available for KPI engineering and dashboard development.

5.1 Integration Approach

The project uses a logical Galaxy Schema to organize the analytical data.

This approach is appropriate for MedTrack because the datasets operate at different levels of detail, or grains:

- Patient-level data represents individual patient records.
- Weekly service data represents aggregated hospital/service activity for a particular period.
- Staff data represents individual staff information.
- Staff schedule data represents staff allocation or scheduling information.
- Readmission data represents patient readmission-related information.

Because these datasets represent different business processes and grains, directly joining all of them together can introduce duplicate records and distort measures such as patient counts, admissions, or resource utilization. Therefore, the integration approach keeps the different analytical areas logically separated while providing the relationships required for analysis.

5.2 Business Keys and Relationships

The project uses relevant business fields to establish relationships between datasets where required.

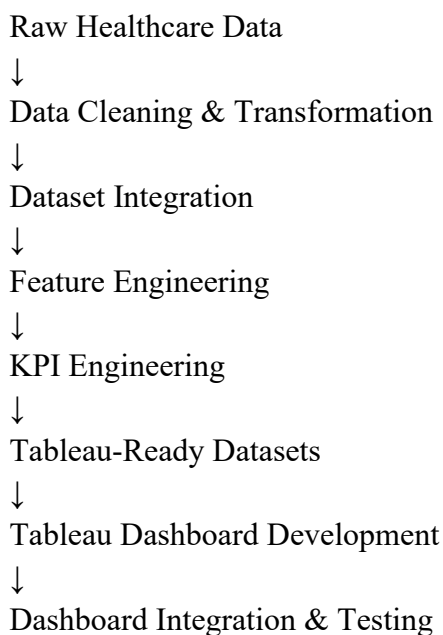
Important documented fields include:

- Service: connects departmental/service-level information with corresponding hospital operations.
- Staff_ID: identifies individual staff members and supports staff-related analysis.
- Week: provides the time-based relationship required for weekly operational and staffing analysis.
- Patient ID: identifies individual patients in patient-level analysis where applicable.

These fields provide the basis for relating information without unnecessarily flattening datasets that have different levels of detail.

5.3 Data Architecture

The overall architecture follows a sequential analytical pipeline:



Each stage produces information required by the next stage.

5.4 KPI Engineering Layer

After the datasets are cleaned and integrated appropriately, the required healthcare KPIs are engineered.

The project specification identifies the following core KPIs:

- Total Admissions
- Occupancy Rate
- Average Length of Stay
- Readmission Rate
- Bed Utilization Rate
- Department Efficiency Score

The KPI-engineering stage produces the final analytical dataset used by Tableau. The documented deliverables for this stage include `hospital_final_dataset.xlsx` and `generate_hospital_kpis.py`.

5.5 Tableau Visualization Layer

The Tableau layer converts the engineered healthcare metrics into interactive visualizations. The final solution consists of four interconnected analytical dashboards:

Hospital Overview

Provides an overall view of hospital performance through major KPIs and supporting operational visualizations.

Patient Flow

Analyzes admissions, discharges, patient volume, Length of Stay, and patient-flow patterns.

Department Analytics

Compares departments using efficiency, occupancy, utilization, and readmission indicators.

Resource Utilization

Analyzes bed capacity, occupancy, utilization, and department-level resource performance.

The original project specification defines these four dashboard areas and requires their integration through global filters, navigation controls, parameter actions, and dashboard linking.

5.6 Dashboard Integration

The four dashboards are designed as a single dashboard suite rather than four independent Tableau reports.

Navigation controls allow users to move between:

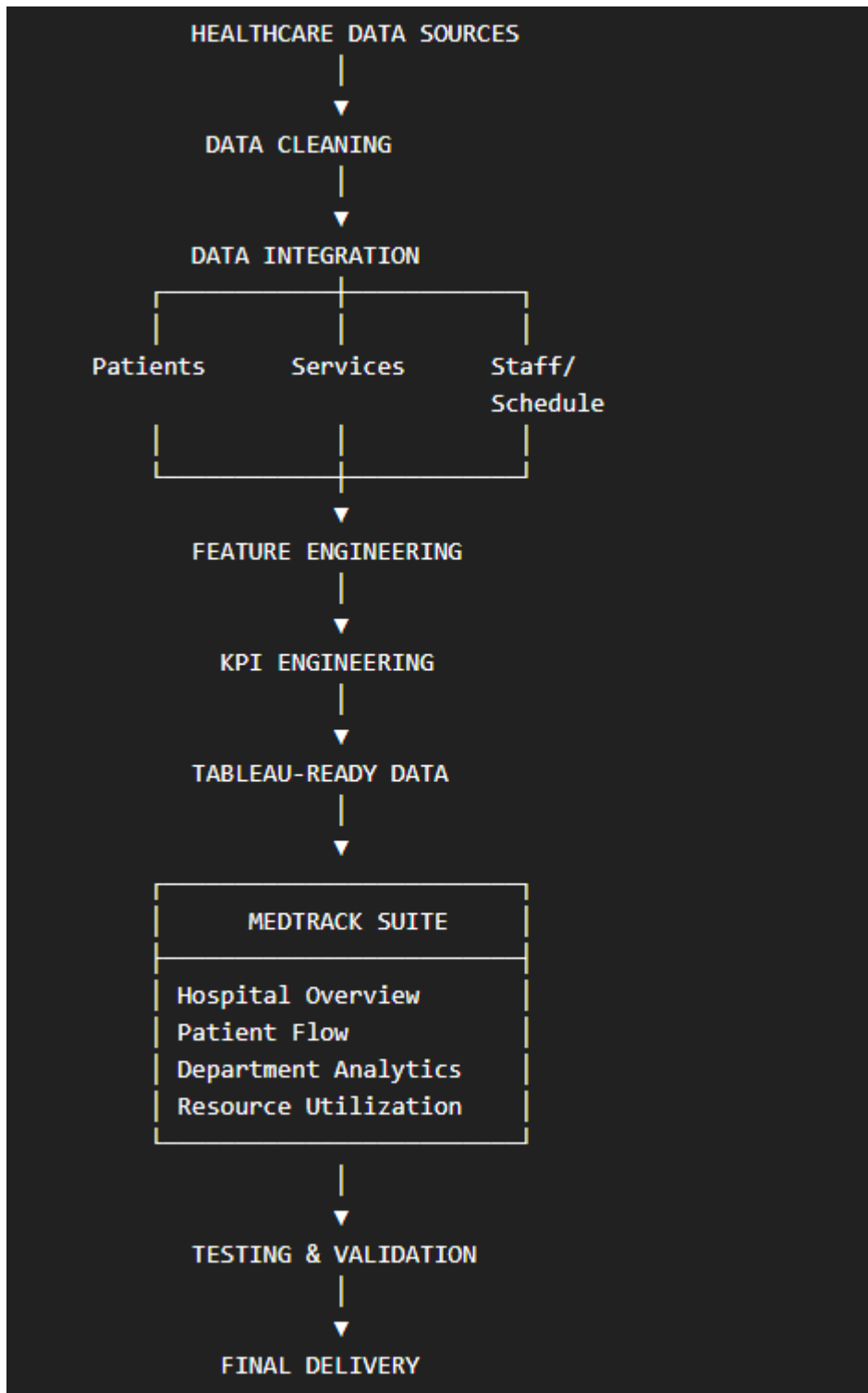
Hospital Overview → Patient Flow → Department Analytics → Resource Utilization

and return to the overview when required.

This integration provides a consistent analytical experience and allows users to move from an overall hospital-level view to more detailed patient, departmental, and resource-level analysis.

5.7 Final Architecture

The complete MedTrack architecture can therefore be summarized as:



The documented project workflow follows the same overall progression from healthcare data collection through cleaning, KPI engineering, dashboard development, integration, testing, and final documentation.

6. KPI Definitions

Key Performance Indicators (KPIs) form the analytical foundation of the MedTrack dashboard suite. They convert the underlying hospital operational data into measurable indicators that can be used to monitor patient activity, hospital capacity, departmental performance, and resource utilization.

The KPIs documented below represent the definitions and calculation logic implemented in this project. They are project-specific analytical definitions and should not automatically be interpreted as universal clinical or hospital reporting standards.

6.1 KPI Reference Table

KPI	Definition	Calculation / Implementation	Unit
Total Patients	Total patient records represented in the analytical dataset.	Patient-count field or distinct Patient ID, where applicable.	Patients
Average Length of Stay (LOS)	Average amount of time patients remain in the hospital between admission and discharge.	Average (Departure Date – Arrival Date)	Days
Occupancy Rate	Proportion of total bed capacity represented as occupied.	$(\text{Total Beds} - \text{Free/Available Beds}) / \text{Total Beds} \times 100$	%
Bed Utilization Rate	Project-specific indicator representing patient admissions relative to total bed capacity.	$\text{Patients Admitted} / \text{Total Beds} \times 100$	%
Readmission Rate	Proportion of patients identified as readmitted within the analytical dataset.	$\text{Readmitted Patients} / \text{Total Patients} \times 100$	%
Department Efficiency Score	Comparative indicator used to evaluate the operational efficiency of hospital departments.	Composite KPI based on the component formula/weighting implemented during KPI engineering.	Score / % as displayed

6.2 Total Patients

Purpose:

Measures the total number of patients represented in the analytical dataset.

Implementation:

Total Patients = Patient count / Distinct Patient ID where applicable

This KPI provides the overall patient-volume context for the other hospital performance indicators.

6.3 Average Length of Stay

Purpose:

Measures the average duration for which patients remain in the hospital.

Implementation:

Average LOS = Average(Departure Date – Arrival Date)

The resulting value is expressed in days.

Average LOS is used throughout the dashboards to compare patient stay patterns between departments and across different time periods.

6.4 Occupancy Rate

Purpose:

Measures the proportion of hospital bed capacity represented as occupied.

Implementation:

Occupancy Rate =
$$(\text{Total Beds} - \text{Free/Available Beds}) / \text{Total Beds} \times 100$$

A higher occupancy rate indicates that a larger proportion of the available bed capacity is occupied.

This KPI is primarily used for monitoring hospital capacity and demand.

6.5 Bed Utilization Rate

Purpose:

Measures the relationship between patient admissions and total bed capacity using the project-specific KPI definition.

Implementation:

$$\text{Bed Utilization Rate} = \frac{\text{Patients Admitted}}{\text{Total Beds}} \times 100$$

The result is expressed as a percentage.

Important project note:

This is the implementation definition used in the MedTrack analytical dataset. It should be described as a project-specific bed-utilization indicator rather than claimed to be a universal hospital-standard formula.

This distinction is particularly important when interpreting why the Bed Utilization Rate and Occupancy Rate may have different values.

6.6 Readmission Rate

Purpose:

Measures the proportion of patients identified as readmitted in the analytical dataset.

Implementation:

$$\text{Readmission Rate} = \frac{\text{Readmitted Patients}}{\text{Total Patients}} \times 100$$

This KPI is used to identify differences in readmission patterns across departments and within the hospital.

It is an analytical monitoring indicator in this project and should not be interpreted as a clinical diagnosis or assessment of individual patient care.

6.7 Department Efficiency Score

Purpose:

Provides a comparative measure for evaluating the operational efficiency of different hospital departments.

The Department Efficiency Score is treated as a composite departmental KPI rather than a single raw measure.

The exact component variables, weighting, and calculation formula should be taken directly from the final KPI-generation implementation used to produce the dataset.

Documentation requirement: Insert the exact formula from `generate_hospital_kpis.py` or the final KPI-engineering workbook before submitting the documentation.

This is the one KPI we should not guess. If an external asks, *"How exactly did you calculate Department Efficiency Score?"*, you need to be able to show the actual formula, not a plausible-looking formula we invented afterward.

6.8 KPI Interpretation

The KPIs should be interpreted together rather than individually.

For example:

- Occupancy Rate provides information about bed-capacity occupancy.
- Bed Utilization Rate provides the project-specific admissions-to-bed-capacity indicator.
- Average LOS provides information about the duration of patient stays.
- Readmission Rate provides an outcome-related operational indicator.
- Department Efficiency Score allows comparative departmental assessment.

Therefore, a single KPI should not be used to conclude that a department is performing well or poorly. The dashboards provide multiple indicators so that hospital operations can be evaluated from several perspectives.

6.9 KPI Validation

Before final delivery, each KPI should be checked against its source data and calculation logic. The project specification requires KPI validation as part of the testing stage, with a target KPI accuracy above 95%.

The validation process should include:

1. Verify the source fields used in each calculation.
2. Recalculate sample KPI values independently.
3. Compare calculated values with the Tableau values.
4. Check percentage and decimal formatting.
5. Check department-level results.
6. Check that filters do not produce unexpected KPI changes.
7. Document any project-specific assumptions.

This makes the KPI layer traceable from source data → calculation → final dashboard value.

7. Dashboard Guide

The MedTrack dashboard suite consists of four interconnected dashboards. Each dashboard addresses a different aspect of hospital operations while using a consistent visual language and KPI framework. Together, they allow users to move from a high-level hospital view to detailed analysis of patient flow, departmental performance, and resource utilization.

7.1 Dashboard 1: Hospital Overview

Purpose:

The Hospital Overview dashboard provides an executive-level snapshot of overall hospital performance. It is designed as the primary entry point to the MedTrack dashboard suite, allowing users to understand the hospital's key operational indicators before moving into more detailed analysis.

Key KPIs:

- Total Patients
- Occupancy Rate
- Bed Utilization Rate
- Average Length of Stay
- Readmission Rate
- Department Efficiency Score

Key Visualizations:

- KPI Cards: Provide an immediate view of the hospital's major performance indicators.
- Admission Share Donut Chart: Shows the distribution of patients/admissions across services or departments.
- Department Efficiency Comparison: Compares the efficiency performance of individual departments.
- Readmission Comparison: Allows readmission performance to be examined across departments.
- Average LOS Analysis: Shows differences in patient stay duration.
- Occupancy Analysis: Provides a visual comparison of occupancy across services.
- Bed Utilization Analysis: Displays the hospital's bed-utilization performance.

Analytical Purpose:

The dashboard answers the question:

"What is the current overall operational performance of the hospital?"

It acts as the starting point for management-level analysis and provides the context required before examining individual operational areas.

7.2 Dashboard 2: Patient Flow

Purpose:

The Patient Flow dashboard focuses on the movement and volume of patients throughout hospital operations. It examines admission and discharge patterns, departmental patient volume, Length of Stay, and patient refusal patterns.

Key Visualizations:

- Monthly Admission Trend: Shows how patient admissions change over time.
- Monthly Discharge Trend: Tracks discharge activity across months.
- Patients by Department: Shows the distribution of patient volume across hospital departments.
- Average LOS Heatmap: Compares Average Length of Stay across departments and months.
- Admitted vs Refused Analysis: Compares the number of patients admitted with patients who were refused.

Analytical Purpose:

The dashboard answers questions such as:

"How is patient volume changing over time?"

"Which departments handle the highest patient volume?"

"Where do longer patient stays occur?"

"How does admitted patient volume compare with refused patient volume?"

This dashboard therefore provides a more detailed view of patient demand and movement after the overall hospital situation has been established through Dashboard 1.

7.3 Dashboard 3: Department Analytics

Purpose:

The Department Analytics dashboard focuses on comparing the operational performance of individual hospital departments.

Instead of looking at the hospital as a single unit, this dashboard allows users to identify differences between services and determine where operational performance varies.

Key KPIs:

- Service Count
- Department Efficiency
- Average Occupancy Rate

Key Visualizations:

- Department Efficiency Comparison: Compares efficiency scores across departments.
- Occupancy Analysis: Shows department occupancy and its relationship to the defined performance/target level.
- Bed Utilization Analysis: Compares bed utilization between departments.
- Readmission Analysis: Examines differences in readmission performance across services.

Analytical Purpose:

The dashboard answers:

"Which departments are performing differently, and how do their operational indicators compare?"

It enables users to examine efficiency, occupancy, utilization, and readmission together rather than relying on a single performance metric.

7.4 Dashboard 4: Resource Analysis

Purpose:

The Resource Analysis dashboard focuses on the utilization of hospital bed resources and the relationship between resource-related indicators and departmental performance.

Key KPIs:

- Average Bed Utilization
- Total Beds
- Average Occupancy

Key Visualizations:

- Bed Utilization Analysis: Examines bed utilization across departments.
- Occupancy Comparison: Compares occupancy levels between departments.
- Resource Summary: Provides exact department-level values for key resource indicators.
- Occupancy vs Department Efficiency: Examines the relationship between occupancy and departmental efficiency.

Analytical Purpose:

The dashboard answers:

"How are hospital bed resources being utilized?"

and:

"How does resource utilization relate to departmental efficiency?"

This allows users to identify differences in capacity usage and examine whether departments with different occupancy levels also demonstrate different efficiency levels.

7.5 Relationship Between the Four Dashboards

The dashboards are designed to provide a progressive analytical journey:

Hospital Overview



"What is happening across the hospital?"



Patient Flow



"How are patients moving through the hospital?"



Department Analytics



"How do individual departments perform?"



Resource Analysis



"How are hospital resources being utilized?"

This structure allows the user to move from high-level monitoring → patient-level operational patterns → departmental comparison → resource analysis.

The four dashboards are intended to function as a unified Tableau dashboard suite, with navigation between dashboards rather than requiring users to treat each dashboard as an independent report. The original project specification identifies the four dashboard areas and requires dashboard integration through navigation controls, filters, parameter actions, and dashboard linking.

7.6 Intended Users

The dashboards are primarily intended for users who need an operational view of hospital performance, including:

- Hospital administrators
- Healthcare operations managers
- Department managers
- Resource planning teams
- Data analysts

The dashboards provide operational and analytical insights and are not intended to replace clinical judgment or serve as a clinical diagnosis system.

8. Dashboard Interaction Guide

The MedTrack dashboard suite is designed to provide an interactive experience rather than presenting static charts. Users can filter departments, explore detailed values through tooltips, and navigate between the four dashboards. Consistent interaction behavior is maintained across the dashboard suite to make the system easier to use and interpret.

8.1 Department and Service Filters

Department or service filters allow users to focus the analysis on a specific hospital department or compare selected departments.

Depending on the dashboard, users can select:

- A single department
- Multiple departments
- All departments

When a department is selected, the relevant charts and KPI values should update according to the configured dashboard interaction.

For example, selecting **Emergency** allows the user to examine Emergency-specific occupancy, utilization, efficiency, patient volume, or other applicable metrics.

8.2 Dashboard Actions

Dashboard actions are used to connect visualizations and allow users to interact with the data.

When an interactive selection is made, the dashboard should update only the worksheets that are intended to respond to that selection.

During testing, each dashboard action should therefore be checked to confirm:

1. The correct worksheet responds to the selection.
2. Unrelated visualizations are not unintentionally filtered.
3. KPI cards update correctly where applicable.
4. The selected department remains consistent across the relevant views.
5. Clearing the selection restores the original dashboard state.

8.3 Tooltips

Tooltips provide additional information when the user places the cursor over a data point, bar, circle, department, or other visualization element.

Where applicable, tooltips should expose:

- Department/service name

- KPI value
- Relevant date or month
- Patient count
- Occupancy
- Bed utilization
- Efficiency
- Readmission rate
- Other information relevant to the selected visualization

Tooltips allow the dashboards to remain visually clean while still providing detailed numerical information when required.

8.4 Dashboard Navigation

The four dashboards are intended to function as one integrated dashboard suite.

Navigation buttons can be provided consistently across the dashboards:

Hospital Overview | Patient Flow | Department Analytics | Resource Analysis

Selecting a navigation button takes the user directly to the corresponding dashboard.

A **Home/Overview** button can also be provided so that users can return to the Hospital Overview from any dashboard.

This navigation structure eliminates the need for users to manually locate individual worksheets and makes the Tableau workbook behave more like a unified analytical application.

8.5 Consistent Dashboard Design

To provide a consistent user experience, the four dashboards should maintain the same:

- Dashboard canvas dimensions
- Background
- Typography
- Title placement
- Color palette
- KPI-card design
- Filter placement

- Navigation placement
- Spacing and alignment conventions

This consistency is particularly important because the four dashboards are presented as components of the same MedTrack solution rather than as independent reports.

8.6 Legend and Filter Management

Legends should be used only when they provide meaningful information for interpreting a visualization.

If a **Service/Department filter** already provides the required selection functionality, a redundant service legend should be removed where appropriate.

This helps:

- Reduce visual clutter
- Create more dashboard space
- Improve readability
- Make the primary interaction mechanism clear

8.7 Interaction Testing

Before final delivery, dashboard interactions should be tested systematically.

The following checks should be performed:

Test	Expected Result
Select a department	Relevant visualizations update
Clear department selection	Dashboard returns to original state
Select multiple departments	Selected departments are reflected correctly
Hover over a visualization	Relevant tooltip appears
Navigate to another dashboard	Correct dashboard opens
Return to Overview	Hospital Overview opens correctly
Apply filters	KPI values and visualizations remain consistent
Reset filters	Original dashboard state is restored

The project specification explicitly includes testing dashboard interactions and validating patient-flow analytics as part of the testing and validation stage.

8.8 Final Interaction Objective

The overall objective is to ensure that users can **filter, explore, interpret, and navigate the MedTrack dashboard suite without confusion**, while maintaining consistent behavior and presentation across all four dashboards.10. KPI Interpretation and Important Definitions

Total Patients: patient count represented by the analytical patient dataset.

Average Length of Stay: average number of days between arrival/admission and departure/discharge.

Occupancy Rate: $(\text{Total Beds} - \text{Free/Available Beds}) / \text{Total Beds} \times 100$.

Bed Utilization Rate: $\text{Patients Admitted} / \text{Total Beds} \times 100$. This is the project-specific implementation definition used in the engineered dataset.

Readmission Rate: $\text{Readmitted Patients} / \text{Total Patients} \times 100$.

Department Efficiency Score: department-level efficiency KPI used for comparison. The exact component weighting/formula must be taken from the final KPI-generation script before submission; it should not be invented in the documentation.

Occupancy Rate and Bed Utilization Rate are separate project KPIs. Because they use different measures/denominators, their values do not have to be identical.

9. Healthcare Operations Methodology

The MedTrack methodology is designed to analyze hospital operations from four complementary perspectives: **patient flow, departmental performance, resource utilization, and readmission patterns**. These perspectives are connected through the KPIs and visualizations implemented in the Tableau dashboard suite.

9.1 Patient Flow Analysis

Patient flow analysis focuses on the movement of patients through the hospital operational process.

The methodology considers:

- **Admissions** as the entry point into the hospital workflow.
- **Discharges** as the exit point from the hospital workflow.
- **Length of Stay (LOS)** as the duration between patient admission and discharge.
- **Monthly trends** to identify changes in patient demand and discharge activity.
- **Department-level patient volume** to identify which services handle greater numbers of patients.

The Patient Flow dashboard uses admission and discharge trends, department-level patient volume, and LOS analysis to provide a visual representation of patient activity over time.

The primary analytical questions are:

How is patient demand changing over time?

Which departments handle the greatest patient volume?

How does patient stay duration vary between departments and months?

9.2 Department Performance Analysis

Department performance is evaluated by comparing multiple operational indicators rather than relying on a single KPI.

The methodology considers:

- Department efficiency
- Occupancy rate
- Bed utilization rate
- Average Length of Stay
- Patient/service volume
- Readmission rate

These indicators provide different perspectives on departmental operations.

For example, a department may have high occupancy but lower efficiency, while another may have lower occupancy and higher efficiency. Therefore, **no single metric is treated as a complete measure of departmental performance.**

The Department Analytics dashboard brings these indicators together to allow comparative analysis between services.

The primary analytical question is:

"How do hospital departments differ in their operational performance?"

9.3 Resource Utilization Analysis

Resource utilization analysis focuses primarily on hospital bed capacity and its relationship with operational demand.

The methodology examines resource-related indicators at both:

- **Hospital level**
- **Department level**

The analysis considers:

- Total bed capacity

- Bed utilization rate
- Occupancy rate
- Department-level resource usage
- Department efficiency

Comparing these indicators helps identify differences in capacity usage between departments and provides a basis for understanding operational demand.

The Resource Analysis dashboard extends this analysis by examining the relationship between **occupancy and department efficiency**, allowing users to identify departments with different combinations of resource usage and operational performance.

9.4 Readmission Analysis

Readmission is treated as an **operational outcome indicator** within the MedTrack project.

The analysis focuses on:

- Overall readmission rate
- Readmission differences between departments
- Patterns that may require further operational investigation

The Readmission Rate is calculated using the project's defined KPI methodology:

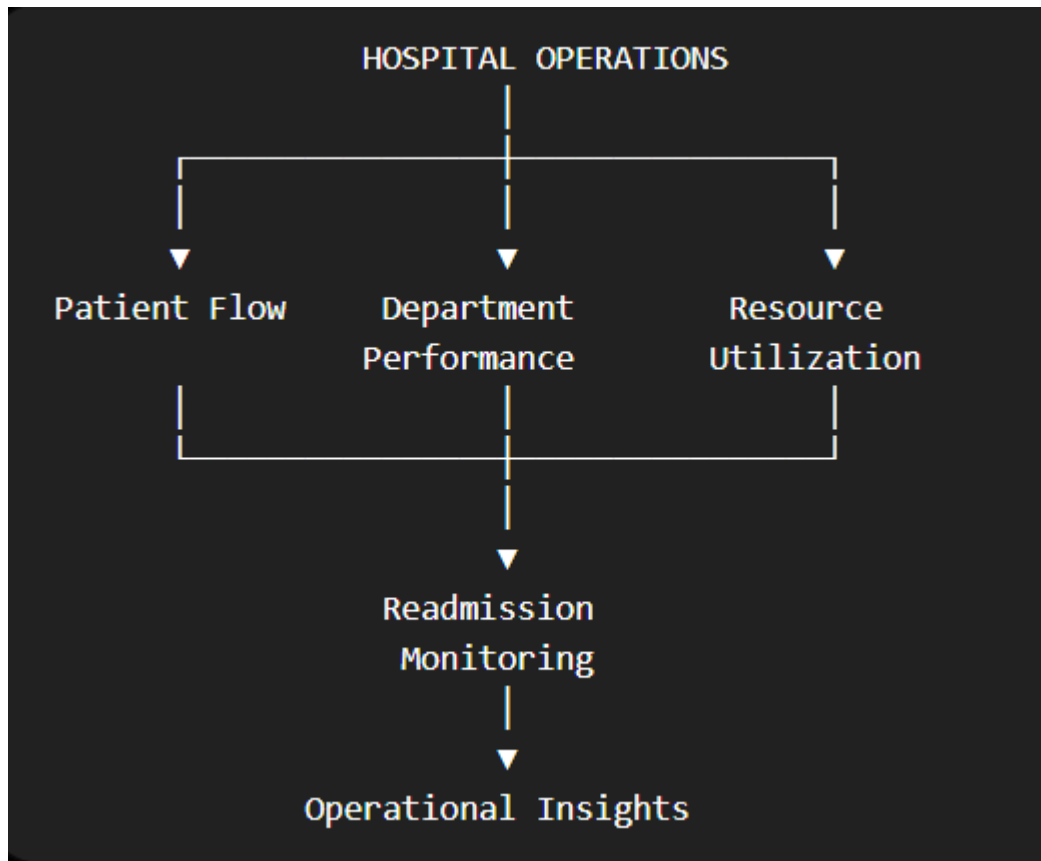
Readmission Rate = Readmitted Patients / Total Patients × 100

Readmission analysis is used to identify patterns in the available dataset and to support operational monitoring.

It is important to clarify that the MedTrack dashboard is **not a clinical diagnosis or patient-treatment system**. A higher or lower readmission value should not, by itself, be interpreted as proof of poor or good clinical care. It is an analytical indicator that may require further investigation using additional clinical and operational information.

9.5 Integrated Healthcare Operations View

The four methodological areas are connected throughout the MedTrack dashboard suite:



This approach ensures that the dashboard does not evaluate hospital performance using a single indicator. Instead, it provides a **multi-dimensional view of hospital operations**, allowing patient activity, departmental performance, resource usage, and readmission patterns to be examined together.

The methodology follows the project's documented scope, which includes patient-flow analysis, department efficiency monitoring, resource-utilization tracking, and hospital performance KPI development.

10. KPI Interpretation and Important Definitions

The KPIs used in MedTrack should be interpreted according to the **specific calculation definitions implemented in this project**. The values shown in the dashboards are intended to provide an operational view of hospital performance and should be interpreted together rather than in isolation.

10.1 Total Patients

Interpretation:

Total Patients represents the number of patient records included in the analytical dataset.

A higher patient count indicates greater patient volume within the analyzed dataset, but it does not by itself indicate higher or lower hospital performance.

Important consideration:

Patient count should be interpreted together with bed capacity, occupancy, Length of Stay, and departmental workload.

10.2 Average Length of Stay

Interpretation:

Average Length of Stay (LOS) represents the average duration that patients remain in the hospital between admission and discharge.

A higher LOS may indicate that patients are staying longer, while a lower LOS indicates shorter average stays.

However, **a higher LOS should not automatically be interpreted as poor performance**. Patient complexity, department type, treatment requirements, and other operational factors can affect LOS.

Unit: Days

10.3 Occupancy Rate

Definition used in MedTrack:

Occupancy Rate = (Total Beds – Free/Available Beds) / Total Beds × 100

Interpretation:

Occupancy Rate represents the proportion of the hospital's total bed capacity that is occupied according to the project's dataset.

For example, an occupancy rate of **90%** indicates that approximately 90% of the defined bed capacity is occupied according to the underlying data.

A high occupancy rate indicates that a large proportion of available capacity is being used.

Important consideration:

Occupancy should be interpreted together with patient volume, Length of Stay, and bed utilization rather than as an isolated measure.

10.4 Bed Utilization Rate

Definition used in MedTrack:

Bed Utilization Rate = Patients Admitted / Total Beds × 100

Interpretation:

This KPI represents the relationship between patient admissions and total bed capacity according to the **project-specific implementation**.

It is important not to describe this calculation as a universal hospital-standard definition. The value is based on the formula implemented during the KPI-engineering stage of this project.

Why can Occupancy and Bed Utilization have different values?

The two KPIs measure different relationships:

Occupancy Rate

Focuses on occupied/free bed capacity.

Bed Utilization Rate

Uses patient admissions relative to total beds.

Therefore, the two percentages **do not need to be equal**.

For example, the current dashboard can show an occupancy rate above 80% while the project-specific bed-utilization indicator is around 50%. This does not automatically indicate a calculation error because the two metrics are calculated differently.

If an external reviewer asks about this difference, the correct response is:

"The two KPIs use different calculation definitions in our project. Occupancy measures occupied bed capacity, while our bed-utilization indicator relates admissions to total beds. Therefore, their values are not expected to be identical."

10.5 Readmission Rate

Definition used in MedTrack:

Readmission Rate = Readmitted Patients / Total Patients × 100

Interpretation:

Readmission Rate represents the proportion of patients identified as readmitted within the available analytical data.

A higher value indicates a greater proportion of identified readmissions in the dataset.

However, the metric should **not be used by itself to conclude that a department provides poor-quality care**. Further clinical and operational investigation would be required to understand the reasons behind a readmission pattern.

10.6 Department Efficiency Score

The Department Efficiency Score is used as a comparative indicator of departmental operational performance.

It allows departments to be compared using the efficiency measure implemented in the KPI-engineering stage.

Important documentation requirement:

The exact component formula and weighting used to calculate this score must be taken directly from the final KPI-generation implementation.

The documentation should **not invent or assume a formula** if it is not present in the final implementation.

10.7 Interpreting KPIs Together

A major principle of MedTrack is that **no single KPI should be used to judge hospital or departmental performance.**

For example:

KPI Combination	Possible Interpretation
High Occupancy + High Patient Volume	Strong demand for hospital services
High Occupancy + High LOS	Capacity may be under pressure because patients remain longer
High Bed Utilization + High Occupancy	Greater use of available bed capacity according to the project's metrics
High Readmission Rate	Requires further investigation into readmission patterns
High Efficiency + High Occupancy	Department is handling high capacity usage while maintaining a strong efficiency score
Low Occupancy + Low Patient Volume	Lower demand or underutilized capacity may warrant investigation

These are **analytical signals, not automatic conclusions**. The dashboard is intended to help users identify patterns that require further investigation.

10.8 Important Data Interpretation Principles

The following principles should be followed when interpreting the MedTrack dashboards:

1. **Use the KPI definition together with the KPI value.**
2. **Do not compare two KPIs as if they measure the same thing.**
3. **Consider department and time-period context.**
4. **Use multiple KPIs to evaluate operational performance.**
5. **Treat unusual patterns as signals for investigation rather than definitive conclusions.**
6. **Do not interpret operational KPIs as clinical diagnoses.**
7. **Remember that the project uses prepared analytical datasets rather than a live hospital information system.**

This ensures that the dashboard insights remain consistent with the methodology and calculation logic used throughout the MedTrack project.

11. Dashboard-Level Insights

The four MedTrack dashboards are designed to provide progressively deeper views of hospital operations. Rather than treating each dashboard as an independent report, the dashboards collectively support analysis from **overall hospital performance to patient flow, departmental performance, and resource utilization**.

11.1 Hospital Overview Insights

The **Hospital Overview** dashboard provides the highest-level view of hospital operations.

It allows users to examine:

- Overall patient volume
- Occupancy performance
- Bed utilization
- Average Length of Stay
- Readmission performance
- Department efficiency
- Distribution of patients across services

The dashboard provides the initial context required to understand the hospital's overall operational position before moving into more detailed analysis.

Key analytical theme:

Overall hospital performance and operational status

11.2 Patient Flow Insights

The **Patient Flow** dashboard focuses on how patient activity changes over time and across departments.

The dashboard highlights:

- Monthly admission patterns
- Monthly discharge patterns
- Patient volume by department
- Average Length of Stay by department and month
- Admitted versus refused patient volume

These visualizations allow users to identify periods of higher or lower patient activity and examine differences in patient stay across departments.

Key analytical themes:

- Patient demand
- Admission and discharge patterns
- Department workload
- Length-of-Stay variation

11.3 Department Analytics Insights

The **Department Analytics** dashboard provides a comparative view of individual hospital departments.

The dashboard allows users to examine differences in:

- Department efficiency
- Occupancy
- Bed utilization
- Readmission rate
- Service volume

This comparison is important because hospital-wide averages can hide differences between individual departments.

For example, two departments may have similar patient volumes but different occupancy, utilization, efficiency, or readmission indicators.

Key analytical theme:**Variation in departmental operational performance*****11.4 Resource Analysis Insights***

The **Resource Analysis** dashboard focuses on hospital bed capacity and resource utilization.

It examines:

- Total bed capacity
- Bed utilization
- Occupancy
- Department-level resource indicators
- Department efficiency
- The relationship between occupancy and efficiency

This dashboard provides a more focused view of how hospital capacity is being used and how resource-related indicators differ between departments.

Key analytical theme:

Resource utilization and capacity-related performance

11.5 Cross-Dashboard Insights

When the four dashboards are considered together, several major analytical themes emerge:

High Occupancy

The dashboards indicate generally high occupancy levels across the hospital and departments. This highlights the importance of monitoring available capacity and patient demand.

Patient Demand and Flow

Admission and discharge trends provide visibility into changes in patient activity over time. Examining both together provides a better understanding of patient-flow patterns than looking at admissions alone.

Departmental Variation

Department-level analysis shows that operational indicators are not uniform across the hospital. Efficiency, occupancy, utilization, LOS, and readmission indicators can vary between services.

Readmission Monitoring

Readmission Rate provides an additional outcome-related indicator that can be used to identify departments or periods that may require further investigation.

Resource Utilization

Bed utilization and occupancy analysis provides visibility into how hospital capacity is being used and supports further investigation of resource-allocation opportunities.

11.6 Overall Analytical Story

The four dashboards form a logical analytical sequence:

Hospital Overview

What is happening across the hospital?

↓

Patient Flow

How are patients moving through the hospital?

↓

Department Analytics

How do individual departments differ?



Resource Analysis

How are hospital resources being utilized?

This structure allows MedTrack to move from **high-level monitoring to detailed operational analysis**, providing a unified view of hospital performance rather than presenting isolated charts. The project specification defines the same four-dashboard structure and identifies patient-flow analysis, department efficiency monitoring, and resource-utilization tracking as core project outcomes.

Important Note

These insights should be presented as **patterns identified in the available dataset**, not as definitive conclusions about hospital quality or clinical performance. Any operational pattern identified through the dashboards should be investigated further using additional hospital and clinical information before making management or clinical decisions.

12. Testing and Validation

Testing and validation were performed as a critical stage of the MedTrack project to ensure that the engineered KPIs, dashboard visualizations, patient-flow analysis, and interactive features function correctly and produce consistent results.

The validation process focuses on two areas: **analytical accuracy** and **dashboard functionality**. Analytical accuracy ensures that the values displayed in Tableau correspond to the underlying data and KPI definitions, while functionality testing ensures that users can interact with and navigate the dashboard suite as intended.

12.1 KPI Calculation Validation

Each KPI is validated against the corresponding source data and its documented calculation logic.

The validation process includes:

- Verifying the source fields used for each KPI.
- Recalculating sample KPI values independently.
- Comparing the independently calculated values with the values stored in the final KPI dataset.
- Checking that Tableau displays the same values as the validated dataset.
- Confirming that percentage-based KPIs are correctly formatted.
- Checking decimal precision and rounding.

The core KPIs subjected to validation include:

- Total Patients
- Average Length of Stay
- Occupancy Rate
- Bed Utilization Rate
- Readmission Rate
- Department Efficiency Score

12.2 Patient and Department Reconciliation

Patient totals and department-level totals are reconciled to identify discrepancies introduced during data cleaning, transformation, or integration.

The checks include:

- Comparing total patient records with the expected patient count.
- Comparing department-level patient counts with the underlying dataset.
- Checking that department totals are not unintentionally duplicated.
- Confirming that the sum of relevant departmental values is consistent with the corresponding hospital-level value where aggregation is logically applicable.
- Checking that filtering by department produces the expected subset of records.

This is particularly important when working with datasets having different levels of detail because inappropriate joins or relationships can cause duplicated records and incorrect aggregate values.

12.3 Date and Length-of-Stay Validation

Date-related fields are validated because they directly affect patient-flow and Length-of-Stay analysis.

The validation includes:

- Checking that arrival/admission dates are stored correctly.
- Checking that departure/discharge dates are stored correctly.
- Verifying that departure dates are not incorrectly earlier than arrival dates where such cases are not expected.
- Recalculating sample Length-of-Stay values manually.
- Comparing the derived LOS values with the values used in Tableau.
- Checking average LOS at hospital and department levels.

This ensures that the Patient Flow dashboard is based on correctly derived time-related measures.

12.4 KPI Range and Formatting Checks

KPI values are checked for reasonable ranges and appropriate presentation.

Examples include:

- Percentage KPIs are displayed as percentages.
- Percentage values are checked for inappropriate negative or excessively large values.
- LOS values are displayed in days.
- Patient counts are displayed as whole-number counts where appropriate.
- Decimal precision is kept consistent across dashboards.
- KPI labels clearly communicate the metric being displayed.

This prevents technically correct values from being presented in a misleading or confusing format.

12.5 Dashboard-to-Source Validation

Values displayed in Tableau are compared with the underlying final datasets.

For selected departments and sample records, the following are checked:

Source Dataset → KPI Dataset → Tableau Worksheet → Dashboard KPI

The purpose is to ensure that no unexpected change occurs between the data-processing stage and the final visualization layer.

This provides traceability from the original analytical data to the value visible to the dashboard user.

12.6 Dashboard Interaction Testing

The interactive functionality of the four dashboards is tested to ensure that user interactions behave as intended.

The following features are tested:

- Department/service filters
- Dashboard actions
- Tooltips
- Navigation buttons
- Dashboard-to-dashboard navigation
- Clearing and resetting filters
- Selection and highlighting behavior

- KPI updates after filtering

For example, when a user selects a particular department, the relevant visualizations should respond according to the configured interaction. Unrelated worksheets should not be unintentionally filtered.

12.7 Patient Flow Validation

The Patient Flow dashboard is specifically validated because it combines multiple operational measures.

The validation includes:

- Comparing admission totals with the source data.
- Comparing discharge totals with the source data.
- Checking monthly admission trends.
- Checking monthly discharge trends.
- Validating department-level patient counts.
- Verifying Average LOS calculations.
- Checking admitted-versus-refused values.

This ensures that the visual patterns shown in the Patient Flow dashboard accurately represent the underlying data.

12.8 Department Comparison Validation

Department Analytics is tested to ensure that departmental comparisons are accurate.

The following are checked:

- Department efficiency values
- Occupancy values
- Bed utilization values
- Readmission values
- Service counts
- Average occupancy KPI
- Department filtering behavior

The purpose is to ensure that differences between departments are based on the actual underlying data rather than inconsistencies introduced during data preparation or visualization.

12.9 Navigation and Dashboard Integration Testing

The four dashboards are tested as an integrated dashboard suite:

Hospital Overview → Patient Flow → Department Analytics → Resource Analysis

Each navigation button should open the correct destination dashboard.

Testing should confirm:

1. Each dashboard can be opened from the navigation bar.
2. Navigation buttons point to the correct dashboard.
3. A Home/Overview button returns the user to the Hospital Overview.
4. Filters and selections behave correctly after navigation.
5. No dashboard objects become inaccessible or distorted after navigation.

The project specification identifies dashboard integration, navigation controls, filters, parameter actions, and dashboard linking as part of the dashboard-development requirements.

12.10 Validation Target

The project specification establishes a **KPI accuracy target above 95%** as part of the testing and validation stage. It also requires the four dashboards to be integrated and their navigation and filters to function correctly.

12.11 Final Validation Outcome

The testing process provides confidence that:

- KPI calculations correspond to their documented definitions.
- Patient and department totals are consistent.
- Length-of-Stay calculations are valid.
- Dashboard values correspond to the final analytical datasets.
- Visualizations represent the intended operational metrics.
- Filters and dashboard actions behave correctly.
- Tooltips provide appropriate information.
- Navigation connects the four dashboards correctly.
- The final Tableau workbook functions as an integrated dashboard suite.

Final validation principle:

Source Data → Validated KPI → Tableau Visualization → Interactive Dashboard

Every major analytical value should be traceable through this sequence before the final workbook is delivered.

13. Assumptions and Limitations

The MedTrack project was developed using prepared healthcare datasets and a defined project scope. The following assumptions and limitations should be considered when interpreting the dashboards and their results.

13.1 Data Source Limitation

The project uses **publicly available healthcare datasets** rather than data obtained from a live hospital information system.

As a result:

- The data does not represent real-time hospital operations.
- The datasets may not contain all information available in a real hospital environment.
- The dashboard reflects the characteristics and limitations of the selected datasets.
- Results should therefore be interpreted as analytical findings from the project dataset rather than as a live representation of hospital performance.

13.2 KPI Definition Limitation

The KPI calculations depend on the structure and fields available in the engineered datasets.

In particular, metrics such as **Occupancy Rate** and **Bed Utilization Rate** use the project-specific calculation definitions documented in Section 6.

Therefore, KPI values should always be interpreted together with their definitions rather than assuming that they represent universal healthcare reporting standards.

13.3 Clinical Interpretation Limitation

MedTrack is an **operational analytics and visualization solution**.

It is not designed to:

- Diagnose patients
- Recommend treatments
- Replace clinical judgment
- Evaluate individual healthcare professionals

- Make direct clinical decisions

The dashboards provide operational patterns and indicators that may support further investigation by appropriate hospital stakeholders.

13.4 Real-Time Data Limitation

The current implementation does not operate on a live hospital information system.

Therefore, the dashboards do not provide:

- Real-time bed availability
- Live patient tracking
- Real-time staff availability
- Live admission/discharge updates
- Real-time hospital alerts

The results represent the data available in the prepared datasets.

13.5 Predictive Analytics Limitation

The current project primarily focuses on **descriptive and diagnostic analytics**.

Advanced capabilities such as:

- Patient-demand forecasting
- Predictive readmission modeling
- Future bed-demand prediction
- Automated resource-allocation recommendations
- Real-time anomaly detection

are outside the current project scope.

These could be considered as future enhancements.

13.6 Longitudinal Data Limitation

The available datasets do not provide the same level of longitudinal information that would typically be available in a comprehensive hospital information system.

Consequently, the dashboards primarily focus on the operational measures available in the project datasets rather than attempting to model complete patient journeys across extended periods.

13.7 Department Efficiency Limitation

The **Department Efficiency Score** is a project-specific composite indicator.

The exact component formula and weighting must be taken directly from the final KPI-generation implementation. The documentation should not introduce a new formula that was not used to generate the final dataset.

Before final submission, the exact formula from `generate_hospital_kpis.py` should be inserted here.

13.8 Dataset Scope

The conclusions generated by MedTrack are limited to the datasets and variables included in the project.

Therefore:

Dashboard patterns should be treated as signals for operational investigation rather than definitive explanations of hospital performance.

14. Project Structure and Delivery

The MedTrack project follows a structured organization so that the datasets, processing scripts, Tableau workbook, documentation, and supporting materials can be maintained and delivered together.

14.1 Recommended Project Structure

MedTrack/

|

|— data/

| |— raw/

| |— cleaned/

| |— final/

|

|— scripts/

| |— data_cleaning.py

| |— generate_hospital_kpis.py

|

|— dashboard/

| |— MedTrack.twbx

| |— dashboard_assets/

```
|
|
|— docs/
|  |— Project_Documentation.pdf
|  |— Project_Documentation.docx
|  |— QA_Checklist.xlsx
|  |— KPI_Definitions.pdf
|  |— Dashboard_Guide.pdf
|
|— README.md
```

14.2 Data Directory

The /data directory contains the datasets used throughout the project.

It should distinguish between:

- **Raw data:** original datasets obtained from the selected sources.
- **Cleaned data:** datasets after data-quality processing.
- **Final data:** KPI-engineered and Tableau-ready datasets.

This separation makes it possible to trace the transformation from the original source data to the final analytical dataset.

14.3 Scripts Directory

The /scripts directory contains the code used for:

- Data cleaning
- Data transformation
- Feature engineering
- KPI calculation
- Final dataset generation

The KPI-generation script should be retained because it provides an important part of the calculation traceability for the final dashboard metrics.

14.4 Dashboard Directory

The /dashboard directory contains:

- Final Tableau workbook
- Dashboard-related assets
- Any supporting visualization resources required for the final presentation

The final workbook should contain the four integrated dashboards:

1. Hospital Overview
2. Patient Flow
3. Department Analytics
4. Resource Analysis / Resource Utilization

14.5 Documentation Directory

The /docs directory contains the formal project documentation, including:

- Dataset documentation
- KPI definitions
- Healthcare operations methodology
- Dashboard guide
- Interaction guide
- Testing and validation documentation
- Assumptions and limitations
- QA checklist

14.6 Final Delivery

The expected project delivery consists of:

- Final Tableau workbook
- Final analytical datasets
- Data-processing and KPI-generation scripts
- Complete project documentation
- QA/testing documentation
- GitHub repository, where required

Tableau Public may additionally be used for dashboard deployment where appropriate.

15. Technical Stack

The MedTrack project uses a combination of data-processing, analytical, visualization, and documentation technologies.

15.1 Python

Python is used for data preparation, transformation, feature engineering, and KPI engineering.

The Python workflow supports:

- Data cleaning
- Data validation
- Data transformation
- Derived-field creation
- KPI calculation
- Final dataset generation

15.2 Pandas and NumPy

Pandas is used for structured data manipulation and analysis, including:

- Reading datasets
- Cleaning records
- Transforming columns
- Handling missing values
- Grouping and aggregation
- Creating analytical fields

NumPy supports numerical operations required during data processing and KPI calculations.

15.3 Excel and CSV

Excel and CSV files are used for:

- Source and intermediate datasets
- Cleaned datasets
- KPI-engineered datasets

- Tableau-ready data
- Data validation and inspection

15.4 Tableau

Tableau Desktop/Tableau Public is used to develop the interactive visualization layer.

Tableau is responsible for:

- KPI cards
- Charts and graphs
- Filters
- Tooltips
- Dashboard actions
- Navigation
- Dashboard integration
- Interactive analysis

15.5 Tableau Dashboard Features

The project uses Tableau functionality such as:

- Filters
- Dashboard actions
- Navigation buttons
- Tooltips
- Parameters where applicable
- Highlighting
- Interactive dashboard linking

These features allow the four dashboards to operate as a unified dashboard suite.

15.6 GitHub and Markdown

GitHub and Markdown are used for project organization and documentation.

The repository can contain:

- Source code

- Dataset documentation
- Project documentation
- README
- Screenshots
- Testing documentation
- Project structure

16. Final Delivery Checklist

The following checklist should be completed before final submission.

Data

- Dataset sources documented.
- Dataset filenames documented.
- Purpose of each dataset documented.
- Data preparation workflow documented.
- Cleaning procedures documented.
- Data integration approach documented.
- Data grain documented.
- Final Tableau-ready dataset prepared.

KPI Engineering

- Total Patients documented.
- Average Length of Stay documented.
- Occupancy Rate documented.
- Bed Utilization Rate documented.
- Readmission Rate documented.
- Department Efficiency Score documented.
- Exact Department Efficiency Score formula verified from the final implementation.
- KPI values validated against the source/final dataset.

Dashboards

- Hospital Overview completed.
- Patient Flow completed.
- Department Analytics completed.
- Resource Analysis completed.
- All four dashboards use consistent dimensions.
- Consistent theme and typography applied.
- KPI cards formatted consistently.
- Filters tested.
- Tooltips tested.
- Dashboard actions tested.
- Navigation buttons added.
- Dashboard-to-dashboard navigation tested.
- Redundant legends removed where appropriate.

Validation

- Patient totals reconciled.
- Department totals reconciled.
- LOS calculations checked.
- KPI ranges checked.
- Percentage formatting checked.
- Dashboard values compared against source data.
- Patient-flow analysis validated.
- Department comparisons validated.
- Interactive functionality tested.
- Final KPI accuracy checked against the project target.

Documentation and Delivery

- Project overview completed.

- Project objectives documented.
- Dataset sources documented.
- Data preparation methodology documented.
- Data integration architecture documented.
- KPI definitions documented.
- Dashboard guide completed.
- Dashboard interaction guide completed.
- Healthcare operations methodology documented.
- KPI interpretation documented.
- Dashboard-level insights documented.
- Testing and validation documented.
- Assumptions and limitations documented.
- Project structure organized.
- Technical stack documented.
- Final documentation placed in /docs.
- Tableau workbook saved and tested.
- GitHub repository organized if required.

17. Conclusion

MedTrack is an integrated **hospital operations and patient analytics dashboard solution** developed to transform prepared healthcare datasets into meaningful operational insights through Tableau.

The project brings together multiple areas of hospital analysis, including **hospital-level performance, patient flow, departmental performance, bed utilization, occupancy, Length of Stay, readmission, and resource utilization**.

The four dashboards provide complementary analytical perspectives:

- **Hospital Overview** provides the executive-level operational picture.
- **Patient Flow** examines admission, discharge, patient volume, and Length-of-Stay patterns.
- **Department Analytics** compares operational performance across departments.

- **Resource Analysis** examines bed capacity, utilization, occupancy, and departmental resource relationships.

The project follows a complete analytical workflow:

Data Collection → Data Preparation → Data Integration → Feature Engineering → KPI Engineering → Tableau Visualization → Dashboard Integration → Testing & Validation → Documentation & Delivery

The resulting solution is intended to help users identify operational patterns and areas requiring further investigation rather than simply displaying raw healthcare data.

The final deliverable therefore consists not only of the Tableau dashboards, but also the **prepared datasets, KPI-engineering process, validation procedures, dashboard interaction design, healthcare operations methodology, assumptions, limitations, and supporting documentation.**

The completed documentation provides traceability from the original datasets through the analytical calculations and finally to the interactive dashboard outputs, making the MedTrack project reproducible, explainable, and suitable for formal project evaluation.

Source Traceability

The documentation has been developed using the project's existing documentation and deliverables as the primary source.

Primary Project Specification

DV_Hospital Operations & Patient Analytics Dashboard

This document defines the project's:

- Problem statement
- Objectives
- Dataset and data workflow
- KPI engineering
- Four-dashboard structure
- Dashboard development
- Testing and validation
- Module 8 documentation requirements
- Project structure

- Final delivery requirements

Dashboard Presentation Specification

Hospital_Dashboard_Presentation_Content.docx

This document provides the planned:

- Dashboard purposes
- KPI structure
- Visualization choices
- Dashboard-level analytical framing

Milestone 4 Presentation Content

Milestone_4_PPT_Content_MedTrack.pdf

This document provides the milestone-level:

- Presentation structure
- Dashboard descriptions
- Key insights
- Challenges
- Future enhancements
- Project conclusion

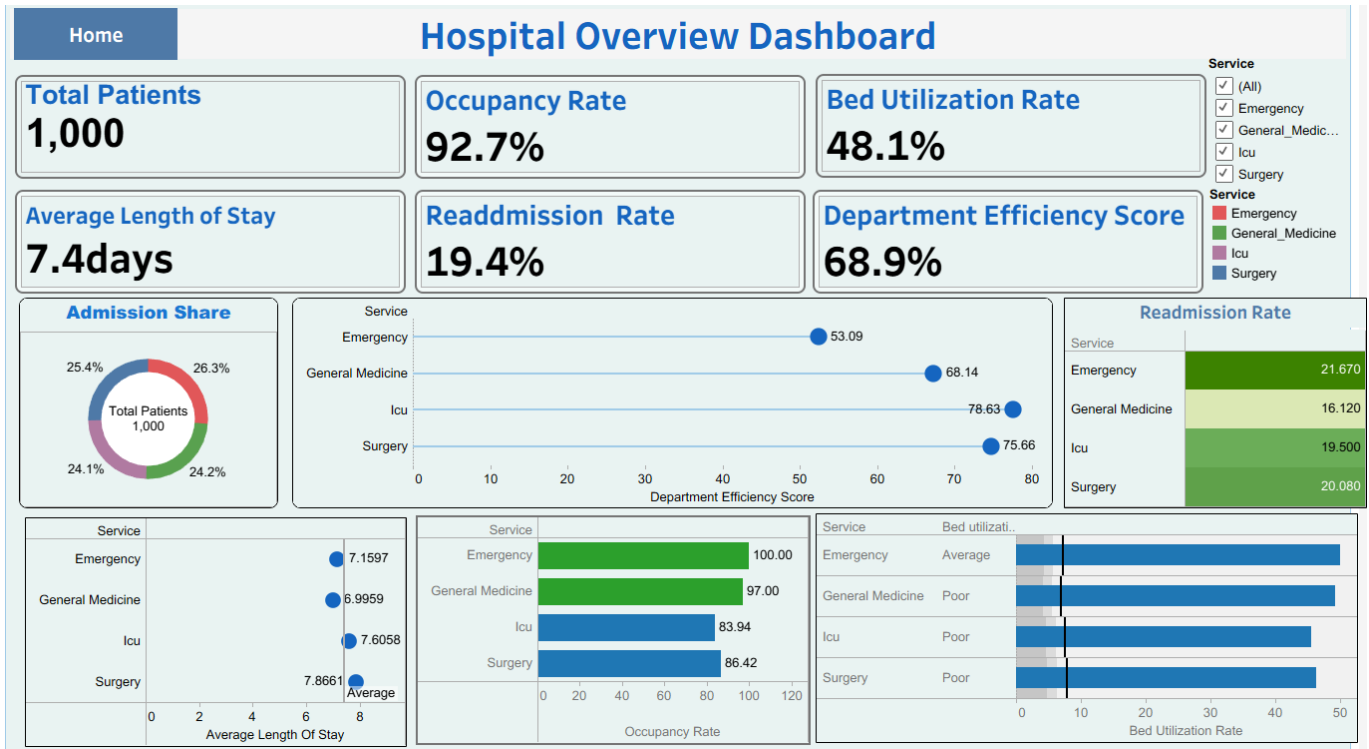
Final Documentation Principle

All KPI formulas, dataset descriptions, and implementation details in the final submission should correspond to the **actual files and implementation used to generate the dashboards**.

In particular, the **Department Efficiency Score formula should be verified against the final KPI-generation script before submission**. That is the one area where we should not guess or retrofit a formula after the fact.

Dashboard:





[Home](#)

Department Analytics

Occupancy Status

Average

Good

Serive Count

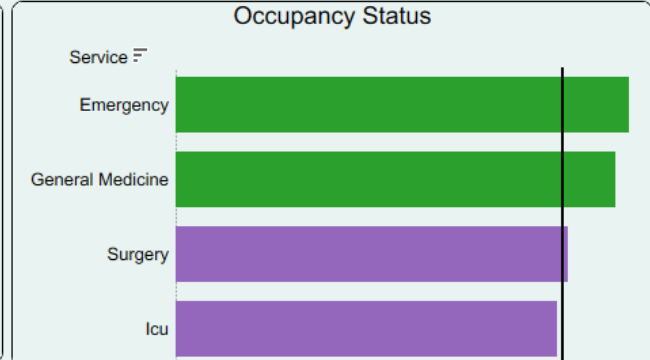
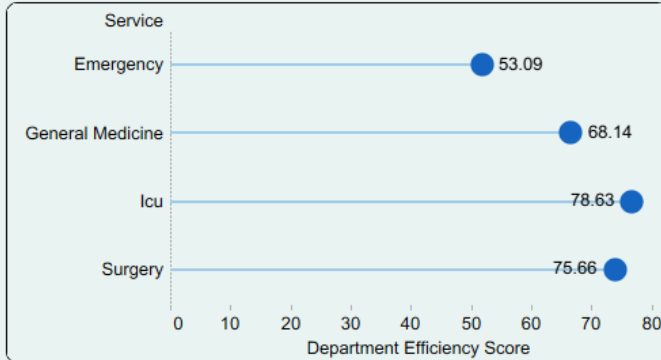
4

Department Efficiency

78.63

Average Occupancy Rate

91.84



Bed Utilization rate

Emergency
50.000Surgery
46.360Icu
45.630General Medicine
49.240

Readmission Rate



[Home](#)

Resource Analysis

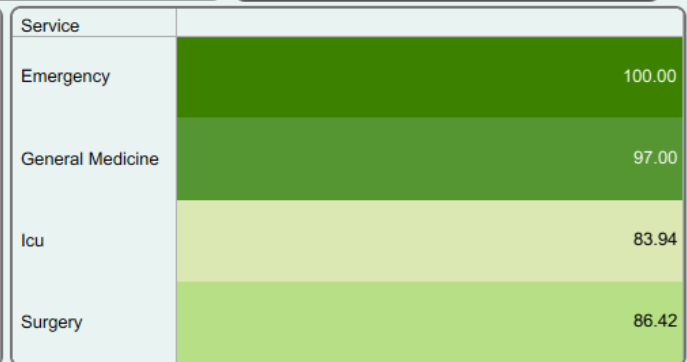
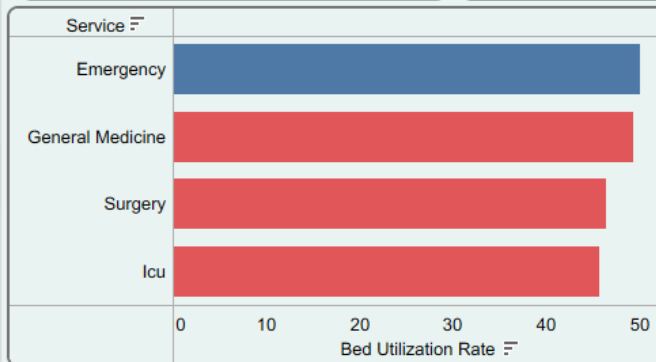
Average Bed Utilization

48.10%

Total Beds

6,312

Average Occupancy rate

91.84%

Resource Summary

Service	Average Length Of St.	Bed Utilization R.	Department Efficiency Sc.	Occupancy Rate
Emergency	7.16	50.00	53.09	100.00
General Medicine	7.00	49.24	68.14	97.00
Icu	7.61	45.63	78.63	83.94
Surgery	7.87	46.36	75.66	86.42

